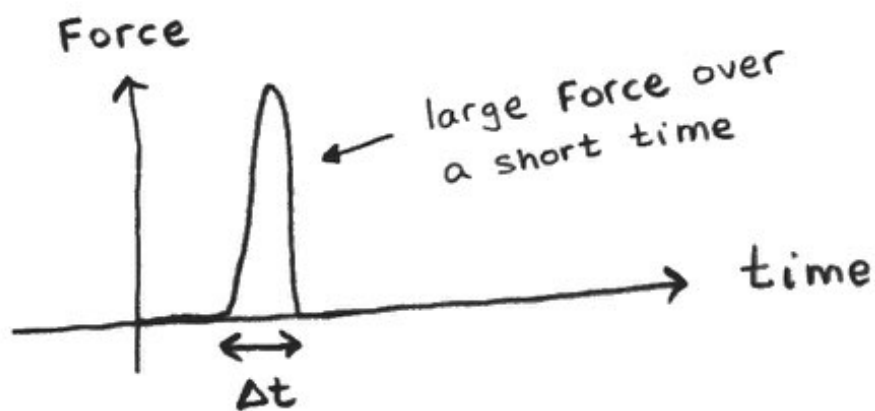


7.5 The Dirac Delta Function

A physical system can involve forces of large magnitude over a short period of time. An electrical circuit can have a very high voltage over a short period of time.

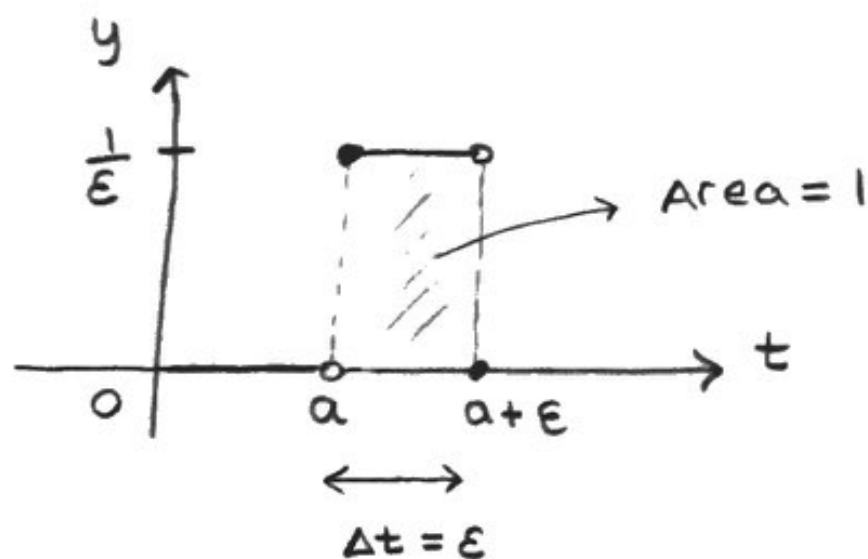
For example, the force on a golf ball when it is hit off the tee might look like:



Dirac Delta can be used to model these forces or Voltages.

Consider the function :

$$\delta_{\epsilon}(t-a) = \begin{cases} \frac{1}{\epsilon} & \text{if } a \leq t < a+\epsilon \\ 0 & \text{otherwise} \end{cases}$$



$\delta_{\epsilon}(t-a)$ is called a unit impulse

since the area under its graph is one.

$$\text{Area} = \int_{-\infty}^{\infty} \delta_{\epsilon}(t-a) dt = 1$$

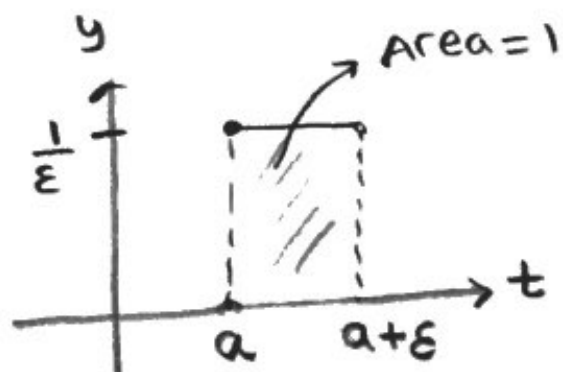
For any $\epsilon > 0$.

The Dirac Delta is defined as :

$$\delta(t-a) = \lim_{\epsilon \rightarrow 0^+} \delta_{\epsilon}(t-a)$$

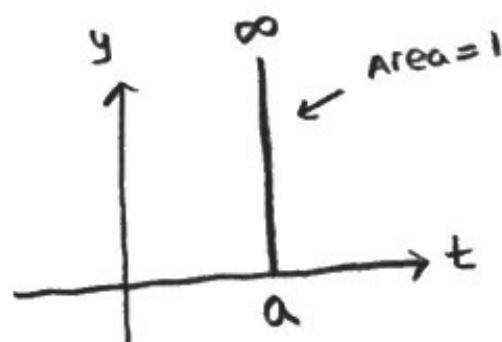
It keeps the property that :

$$\int_{-\infty}^{\infty} \delta(t-a) dt = 1$$



$$y = \delta_{\epsilon}(t-a)$$

limit as
 $\epsilon \rightarrow 0^+$



$$y = \delta(t-a)$$

Dirac Delta

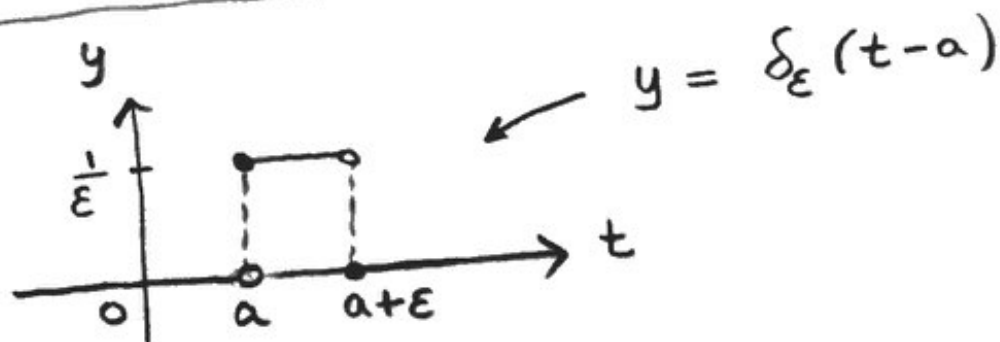
$$\delta(t-a) = \begin{cases} \infty & \text{if } t=a \\ 0 & \text{if } t \neq a \end{cases}$$

Our main focus in this section is to learn how to solve initial-value problems involving Dirac deltas.

To do this we need to know what is the Laplace transform of $\delta(t-a)$.

By definition:

$$\mathcal{L}\{\delta(t-a)\} = \lim_{\epsilon \rightarrow 0^+} \mathcal{L}\{\delta_\epsilon(t-a)\}.$$



observe that:

$$\delta_\epsilon(t-a) = \frac{1}{\epsilon} u(t-a) - \frac{1}{\epsilon} u(t-(a+\epsilon))$$

$$\Rightarrow \mathcal{L}\{\delta_\epsilon(t-a)\} = \frac{1}{\epsilon} \left(\frac{e^{-as}}{s} \right) - \frac{1}{\epsilon} \left(\frac{e^{-(a+\epsilon)s}}{s} \right)$$

$$= \frac{e^{-as}}{s} \left(\frac{1 - e^{-\epsilon s}}{\epsilon} \right)$$

$$\mathcal{L}\{\delta(t-a)\} = \lim_{\epsilon \rightarrow 0^+} \mathcal{L}\{\delta_\epsilon(t-a)\}$$

$$= \lim_{\epsilon \rightarrow 0^+} \frac{e^{-as}}{s} \left(\frac{1 - e^{-\epsilon s}}{\epsilon} \right)$$

The limit has the form $\left(\frac{0}{0}\right)$ so we can use l'Hôpital's rule. Note that ϵ is the variable.

$$\mathcal{L}\{\delta(t-a)\} \stackrel{(H)}{=} \lim_{\epsilon \rightarrow 0^+} \frac{e^{-as}}{\cancel{s}} \left(\frac{\cancel{s} e^{-\epsilon s}}{1} \right)$$

$$= e^{-as} e^0$$

$$= e^{-as}$$

$$\mathcal{L}\{\delta(t-a)\} = e^{-as}$$

An alternative way to derive this would involve the sifting property of $\delta(t-a)$:

$$\int_{-\infty}^{\infty} f(t) \delta(t-a) dt = f(a).$$

$$\mathcal{L}\{\delta(t-a)\} = \int_0^{\infty} e^{-st} \delta(t-a) dt$$

$$= \int_{-\infty}^{\infty} e^{-st} \delta(t-a) dt$$

(since $\delta(t-a) = 0$ over $t < 0$)

$$= \boxed{e^{-as}} \leftarrow \text{By the sifting property.}$$

ex: ① Solve the IVP

$$\begin{cases} x''(t) + 4x(t) = 2\delta(t - \pi) \\ x(0) = 0 \\ x'(0) = 3 \end{cases}$$

Note: This IVP can model a mass-spring system with negligible damping ($\beta \approx 0$). At $t = \pi$, the mass is given a sharp blow.

To solve the IVP, take the Laplace transform on both sides of the DE.

$$\mathcal{L}\{x''(t)\} + 4\mathcal{L}\{x(t)\} = 2\mathcal{L}\{\delta(t - \pi)\}$$

$$\left[s^2 X(s) - s x(0) - x'(0) \right] + 4 X(s) = 2e^{-\pi s}$$

$$(s^2 + 4) X(s) - 3 = 2e^{-\pi s}$$

$$\Rightarrow X(s) = \frac{3}{s^2 + 4} + \frac{2e^{-\pi s}}{s^2 + 4}$$

$$\Rightarrow X(s) = \frac{3}{2} \left(\frac{2}{s^2 + 4} \right) + e^{-\pi s} \left(\frac{2}{s^2 + 4} \right)$$

Recall: $\mathcal{L}^{-1}\{e^{-as} F(s)\} = f(t-a)u(t-a)$

$$x(t) = \mathcal{L}^{-1}\{X(s)\}$$

$$x(t) = \frac{3}{2} \sin(2t) + \sin(2(t-\pi))u(t-\pi)$$

$$\begin{aligned}\sin(a(t-\pi)) &= \sin(at-2\pi) \\ &= \sin(at)\end{aligned}$$

$$\Rightarrow x(t) = \frac{3}{2} \sin(at) + \sin(at) u(t-\pi)$$

or

$$x(t) = \begin{cases} \frac{3}{2} \sin(at) & \text{for } 0 \leq t < \pi \\ \frac{5}{2} \sin(at) & \text{for } t \geq \pi \end{cases}$$

ex: (2) Solve the IVP

$$\begin{cases} y''(t) + 2y'(t) + y(t) = \delta(t-2) \\ y(0) = 1 \\ y'(0) = -1 \end{cases}$$

$$\mathcal{L}\{y''(t)\} + 2\mathcal{L}\{y'(t)\} + \mathcal{L}\{y(t)\} = \mathcal{L}\{\delta(t-2)\}$$

$$[s^2 Y(s) - sy(0) - y'(0)] + 2[sY(s) - y(0)] + Y(s) = e^{-2s}$$

$$(s^2 + 2s + 1)Y(s) - s + 1 - 2 = e^{-2s}$$

$$Y(s) = \frac{s+1}{s^2+2s+1} + \frac{e^{-2s}}{s^2+2s+1}$$

$$Y(s) = \frac{s+1}{(s+1)^2} + \frac{e^{-2s}}{(s+1)^2}$$

$$Y(s) = \frac{1}{s+1} + e^{-as} \left(\frac{1}{(s+1)^2} \right)$$

Recall:

$$\mathcal{L}^{-1}\{e^{-as} F(s)\} = f(t-a) u(t-a)$$

For the second term:

$$F(s) = \frac{1}{(s+1)^2} \quad \text{and } a=2$$

$$\mathcal{L}^{-1}\{F(s)\} = t e^{-t}$$

$$\Rightarrow y(t) = \mathcal{L}^{-1}\{Y(s)\}$$

$$\Rightarrow y(t) = e^{-t} + (t-2)e^{-(t-2)} u(t-2)$$